

WEEK 3 TASK

Analytical Model Development and Validation

Hypothetical Data Science Project: Customer Churn Prediction

Student Name	_____
Internship / Course	Data Science Internship
Task	Week 3 – Analytical Model Development and Validation
Project Type	Classification / Predictive Analytics
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1. Introduction

Analytical model development is the stage of a data science project where prepared data is used to build a predictive model. The model learns patterns from historical data and uses them to make predictions on new data. Model validation is important because a model should perform well not only on training data but also on unseen data.

2. Project Objective

The objective of this hypothetical project is to predict whether a customer is likely to leave a service. This is called **customer churn prediction**. It is a binary classification problem because the target has two outcomes: **Churn = Yes** or **Churn = No**.

3. Dataset Description

The hypothetical dataset contains customer information that may help predict churn.

Feature	Description	Type
Customer Age	Age of customer	Numerical
Monthly Charges	Monthly amount paid	Numerical
Tenure	Months with the service	Numerical
Contract Type	Monthly, yearly, etc.	Categorical
Payment Method	Method used for payment	Categorical
Support Calls	Number of support calls	Numerical
Churn	Whether customer left	Target

4. Criteria for Selecting Suitable Algorithms

The algorithm should be selected according to the type of problem, dataset size, interpretability, computational cost, expected performance, and ability to handle the available features. Since the target is binary, classification algorithms are suitable.

Algorithm	Advantages	Limitations	Role
Logistic Regression	Simple, fast, interpretable	May miss complex patterns	Baseline
Decision Tree	Easy to understand; non-linear patterns	Can overfit	Comparison
Random Forest	Strong performance; robust; complex patterns	Less interpretable	Final model

5. Selected Algorithm: Random Forest

Random Forest is selected as the final algorithm because it combines multiple decision trees to produce a more robust prediction. It can model non-linear relationships and generally works well with tabular classification data. It also reduces the risk of depending on a single decision tree.

6. Model Development Workflow

1. Prepared Dataset
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2. Data Cleaning & Preprocessing

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3. Feature Selection / Encoding
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4. Train-Test Split (80% / 20%)
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5. Train Classification Models
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6. Predict on Test Data
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7. Calculate Validation Metrics
↓
8. Compare Models & Select Final Model

7. Data Splitting

The dataset is divided into two parts: approximately **80% training data** and **20% testing data**. The training set is used to learn patterns, while the test set measures performance on unseen data. Stratified splitting can help maintain a similar class distribution in both sets.

8. Model Training and Testing

During training, customer features and known churn outcomes are provided to the algorithm. The Random Forest learns relationships between customer characteristics and churn. After training, the model predicts churn for the test set. These predictions are compared with the true test labels.

9. Validation Metrics

Metric	Meaning	Why It Matters
Accuracy	Percentage of all predictions that are correct.	Useful when classes are reasonably balanced.
Precision	Of predicted churners, how many actually churned.	Useful when false churn alerts are costly.
Recall	Of actual churners, how many were identified.	Important when missing churners is costly.
F1-Score	Harmonic mean of precision and recall.	Balances false positives and false negatives.

10. Metric Formulas

Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$
F1-Score	$2 \times (Precision \times Recall) / (Precision + Recall)$

TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

11. Confusion Matrix

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	True Negative (TN)	False Positive (FP)
Actual: Churn	False Negative (FN)	True Positive (TP)

12. Hypothetical Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82%	78%	71%	74%
Decision Tree	84%	80%	76%	78%
Random Forest	88%	86%	82%	84%

These are illustrative results for the hypothetical project, not results from a real dataset. Random Forest gives the strongest overall balance in this example.

13. Validation Strategy

The test set should remain separate from training. Performance should be measured using multiple metrics rather than accuracy alone. For a more robust assessment, k-fold cross-validation can be performed on the training data. Hyperparameters can be tuned using validation results, while the final test set is reserved for the final unbiased evaluation.

14. Model Development Checklist

- Verify data quality and target labels.
- Clean missing and inconsistent values.
- Encode categorical variables and scale features where required.
- Split data into training and testing sets.
- Train baseline and candidate models.
- Evaluate accuracy, precision, recall, and F1-score.
- Inspect the confusion matrix.
- Compare models and select the most suitable one.
- Document assumptions, limitations, and results.

15. Conclusion

Analytical model development is a structured process rather than simply choosing an algorithm and training it. For customer churn prediction, Logistic Regression, Decision Tree, and Random Forest are suitable candidates. Random Forest is selected in this plan because it can capture complex patterns and provides strong classification performance. Validation with multiple metrics helps determine whether the model is reliable on unseen data.

16. Future Improvements

Future work can include collecting more representative data, handling class imbalance, tuning model hyperparameters, testing Gradient Boosting and other algorithms, using explainability techniques, monitoring performance after deployment, and retraining the model when customer behavior changes.

17. References

1. Scikit-learn documentation – classification algorithms and model evaluation.
2. General machine learning textbooks/resources on supervised learning and model validation.
3. Python documentation – concepts related to machine learning implementation.